

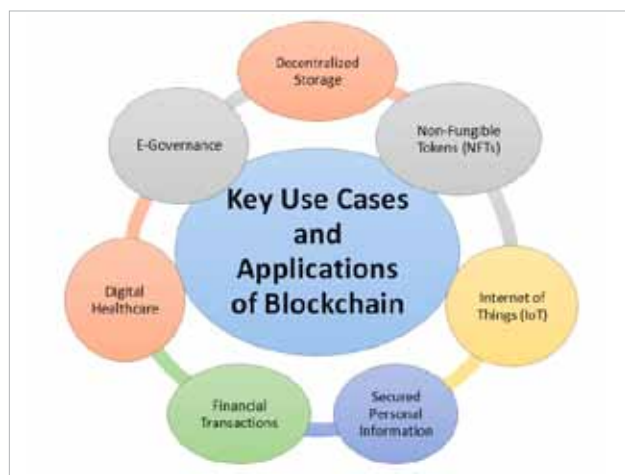


Figure 1: Key use cases and applications of blockchain

potential in different sectors. In supply chain management, for instance, blockchain's transparency and immutability have been harnessed to track and verify the origins and movements of products, thereby enhancing trust and authenticity within supply chains.

Similarly, in healthcare, blockchain technology has been considered for securely managing patient data, ensuring privacy and integrity while enabling authorised access by healthcare providers. In the real estate sector, blockchain's

ability to provide transparent and immutable records of property ownership and transactions could revolutionise the industry, reducing fraud and streamlining the cumbersome processes involved in property transactions.

Crucially, the sphere where blockchain showcases immense promise is in data storage and backup. As industries increasingly grapple with the vulnerabilities of centralised data storage, blockchain's decentralised nature and robust security features offer a compelling solution. The distributed and immutable storage model of blockchain reduces the risk of data breaches, offers enhanced resilience against cyber threats, and provides a more secure and reliable framework for data backup.

Blockchain for decentralised storage

Decentralised storage, powered by blockchain technology, introduces a transformative approach to handling data. This involves breaking down data into smaller, encrypted segments, a process known as sharding. These encrypted fragments are then distributed across a network of nodes in a decentralised manner. This fragmentation and distribution not only fortify the security of the stored data but also augment its accessibility and reliability.

Also, the redundancy created by distributing data across multiple nodes in a decentralised blockchain network ensures that even if one or several nodes fail or are compromised, the data remains accessible and intact due to its availability across other nodes.



Figure 2: IPFS decentralised platform in blockchain